Audio Amplifier Design Report

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Statement of originality

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Abstract

This project presents the design, simulation, implementation and testing of a two-stage audio amplifier intended to amplify the output signal of a mobile phone and drive an 8 Ω speaker load. The amplifier consists of an active filter stage based on the TL071 operational amplifier and a power amplifier stage using the OPA551 operational amplifier. The first stage provides signal amplification and frequency shaping using an active band-pass filter, while the second stage provides the current required to drive the low-impedance speaker. The circuit was designed to operate from a single DC power supply and produce an output signal of approximately 3 Vpp within the audio frequency range. The amplifier was first analysed theoretically and simulated using Proteus before being implemented on a printed circuit board (PCB). Experimental testing showed good agreement with the simulated results, confirming that the amplifier achieved the desired gain and frequency response while operating without distortion.

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1. Introduction

The aim of this project was to design, simulate, construct and test a two-stage audio amplifier capable of amplifying the output signal from a mobile phone and driving an 8 Ω speaker load. The amplifier was required to operate from a single DC power supply and meet specific design specifications, including a defined gain, target output voltage and controlled frequency response within the audio range. The circuit was implemented using two operational amplifier stages: an active filter stage based on the TL071 operational amplifier and a power amplifier stage based on the OPA551 operational amplifier.

The first stage of the amplifier provides signal conditioning and amplification using an active band-pass filter configuration. This stage increases the input signal amplitude while also controlling the frequency response of the circuit by defining the lower and upper cut-off frequencies. The second stage acts as a power amplifier capable of delivering sufficient current to drive the low-impedance speaker load. The use of a dedicated power amplifier stage is necessary because standard operational amplifiers, such as the TL071, cannot supply the current required to drive an 8 Ω load directly. The amplifier operates from a single power supply, requiring the signal to be biased around half of the supply voltage to allow symmetrical signal swing.

The overall design targets include achieving a mid-band gain sufficient to amplify a mobile phone audio signal to approximately 3 Vpp at the speaker output while maintaining stable operation across the desired audio frequency range. The amplifier design was first analysed theoretically and then simulated using circuit simulation software before being implemented on a printed circuit board (PCB). Experimental measurements were performed to characterise the performance of the amplifier and compare the results with the theoretical calculations and simulation results.

This report is organised as follows. Section 2 provides the technical background relevant to operational amplifiers and active filter design. Section 3 describes the amplifier design, calculations and PCB implementation. Section 4 presents the simulation and experimental

results obtained from testing the circuit. Finally, Section 5 summarises the references used throughout the report.

2. Technical Background

Audio amplifiers increase the amplitude of electrical audio signals so they can drive loads such as loudspeakers. Signals from devices like mobile phones are typically low-power voltage signals and cannot supply enough current to drive a speaker directly. Therefore, amplifier circuits are required to increase the signal amplitude and provide sufficient current to the load.

In many audio systems, the frequency response is limited to a specific range using filters. A bandpass filter allows frequencies within a defined range to pass while attenuating frequencies above and below this range. Audio systems typically operate within approximately 20 Hz to 20 kHz, although wider bandwidths are often used to reduce distortion within the audible range.

Active bandpass filters are commonly implemented using operational amplifiers together with resistors and capacitors. The lower cut-off frequency is created by a high-pass RC network, while the upper cut-off frequency is produced by a low-pass RC network. These frequencies correspond to the points where the gain drops by 3 dB.

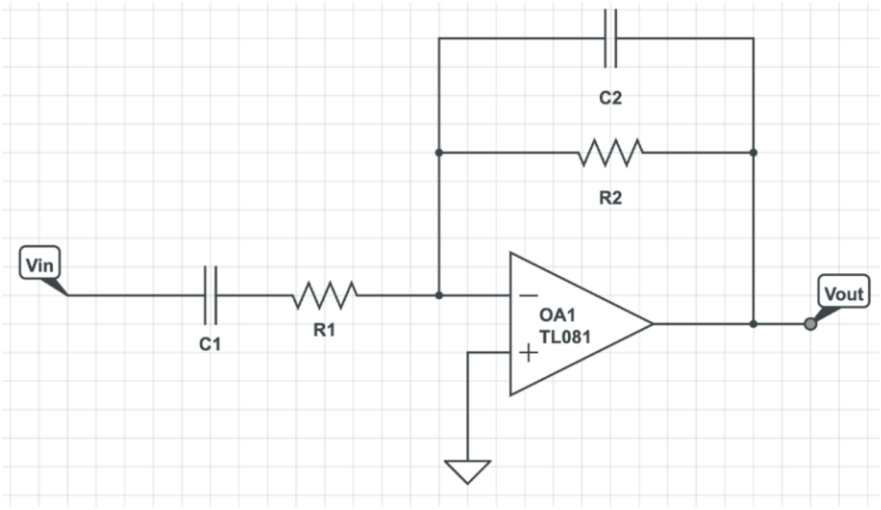


Figure 1 - Generic active bandpass filter using an operational amplifier.

The lower cut-off frequency is given by:

$$\boldsymbol{f_L = \frac{1}{2\pi RC}}$$

Equation 1

The upper cut-off frequency is given by:

$$f_U = \frac{1}{2\pi RC}$$

Equation 2

Operational amplifier circuits also provide voltage gain. For an inverting amplifier configuration, the gain is determined by the resistor ratio:

$$A_v = -\frac{R_f}{R_{in}}$$

Equation 3

Equation 1, Equation 2 and Equation 3 describe the relationship between component values and the frequency response of the active bandpass filter.

Operational amplifiers are widely used in audio circuits due to their high input impedance, low output impedance, and ability to implement precise gain and filtering functions.

2.1 TL071 Operational Amplifier

The TL071 is a JFET-input operational amplifier commonly used in audio applications. The JFET input stage provides high input impedance and low input bias current, which makes the device suitable for amplifying low-level audio signals.

Key characteristics of the TL071 include:

- High input impedance
- Low noise performance
- High slew rate for audio signals
- Wide operating supply voltage range
- Unity gain stability

The device also has a defined gain-bandwidth product, which limits the maximum frequency at which a given gain can be maintained. Due to its limited output current capability, the TL071 is typically used for signal processing and filtering rather than driving speakers directly.

2.2 Power Amplifiers and the OPA551

Loudspeakers typically have low impedances (around 8 Ω), which require significant output current. Small-signal operational amplifiers such as the TL071 cannot supply sufficient current to drive such loads.

The OPA551 is a power operational amplifier designed to drive lower impedance loads. It provides higher output current capability and can operate over a wide supply voltage range.

Important features of the OPA551 include:

- High output current capability
- Wide operating voltage range
- High slew rate
- Unity gain stability

- Built-in protection features

In many circuits, the OPA551 is used as a buffer amplifier, meaning it provides current amplification while maintaining approximately unity voltage gain. This allows the signal from the first stage to drive a speaker without distortion.

2.3 Multi-Stage Amplifier Design

Audio amplifier systems often use multiple stages so that each stage performs a specific function.

The first stage typically performs signal conditioning, including filtering and voltage amplification. In this design, the TL071 is used for this stage due to its low noise characteristics and suitability for active filter circuits.

The second stage is used for power amplification. The OPA551 is used in this stage because it can supply the higher currents required to drive a loudspeaker.

Two TL071 devices cannot be used for both stages because the TL071 cannot provide sufficient output current for a low-impedance load such as a speaker.

2.4 Single Supply Operation

Many audio amplifiers operate from a single supply voltage rather than dual positive and negative supplies. When using a single supply, the signal must be biased to a midpoint voltage so that the AC waveform can swing in both directions without clipping.

This midpoint voltage is generated using a resistor divider network. The bias voltage is typically equal to approximately half of the supply voltage. Coupling capacitors are then used to remove DC components between stages so that only the AC audio signal is amplified.

2.5 Stereo and Mono Signals

Mobile phones normally output **stereo audio signals**, consisting of left and right channels. In small amplifier systems, these signals are often combined into a single **mono signal** using a resistive mixing network. The combined signal can then be amplified and used to drive a single speaker.

3. Design Description

This section describes the design specifications and calculations used to implement the audio amplifier. The design is based on the measured output characteristics of a mobile phone audio source and the required output voltage needed to drive the power amplifier stage. The final design specifications are summarised in Table 1.

3.1 Technical specifications

Table 1 summarises the electrical characteristics and design specifications of the audio amplifier system. The input signal is provided by a mobile phone headphone output and the amplifier is powered using a single 12 V DC supply. The design includes a bandpass response

with defined lower and upper cut-off frequencies to limit the amplified signal to the desired frequency range.

Electrical Characteristic	Value	Notes
Input signal	0.32 Vpp	Mobile phone headphone output measured using 440 Hz tone
Supply voltage	12 V	External DC power supply via terminal block
Lower cut-off frequency	8 Hz	Target and achieved value from RC filter
Upper cut-off frequency	30 kHz	Determined by feedback network
Output power of power amplifier	≈0.14 W	Calculated for 3 Vpp output across 8 Ω load
Resistance of speaker	8 Ω	Standard loudspeaker impedance
Maximum output current of power amplifier	200 mA	OPA551 datasheet limit
Maximum output voltage of power amplifier	≈3 Vpp	Design output amplitude
Measured gain	9.375 (19.4 dB)	Determined from input/output ratio
Controls		
Volume control?	None	Input level controlled by mobile phone
Indicators		
On/off Switch? LED Indicator?	Slide switch Red LED	Disconnects supply voltage Indicates power is present

Figure 2 - Table Electrical Specifications of the Audio Amplifier.

3.2 Overview of the design

The amplifier was designed as a **two-stage audio amplifier** consisting of a signal conditioning stage followed by a power amplification stage. The first stage uses a TL071 operational

amplifier configured as an active bandpass filter and voltage amplifier. This stage performs stereo signal mixing and provides the required voltage gain while limiting the amplifier bandwidth.

The second stage uses an OPA551 power operational amplifier configured as a buffer amplifier. This stage provides the current required to drive the loudspeaker while maintaining the signal amplitude generated by the first stage.

The circuit operates from a **single 12 V supply**, with a bias network used to create a mid-supply reference voltage for the operational amplifier inputs. Coupling capacitors are used between stages to remove DC components while allowing the AC audio signal to pass.

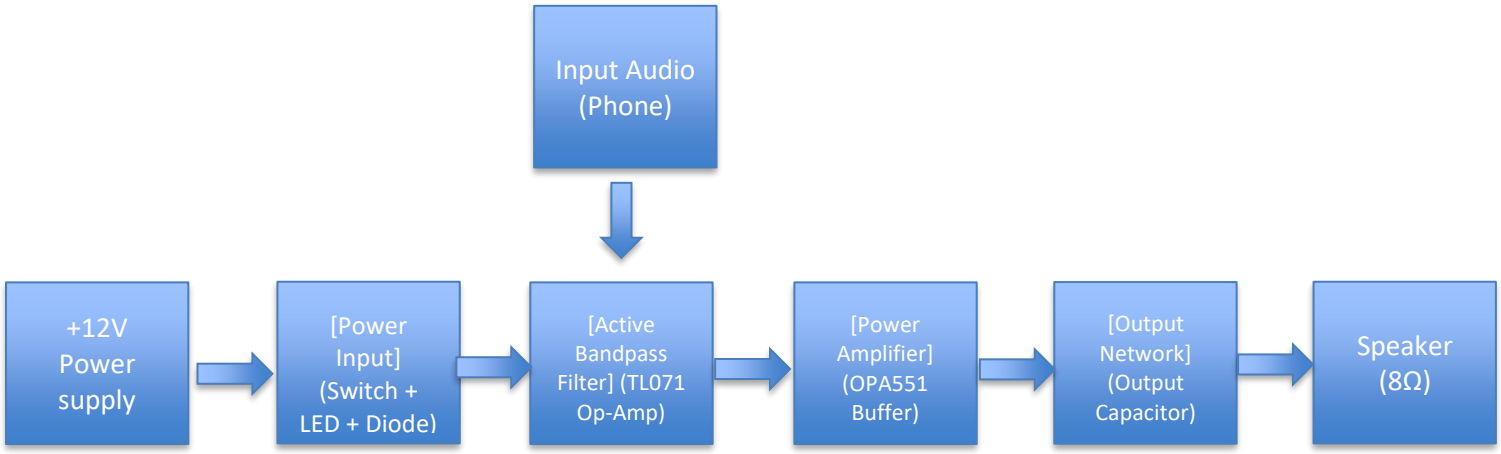


Figure 3 - Top level block diagram.

3.3 Design calculations

The input signal for the amplifier was characterised using a **440 Hz reference tone**, which is commonly used as a standard audio test frequency. The output voltage of the mobile phone was measured at different volume settings to determine the maximum usable input signal.

Figure 2 shows the measured peak-to-peak output voltage of the mobile phone as a function of the volume setting.

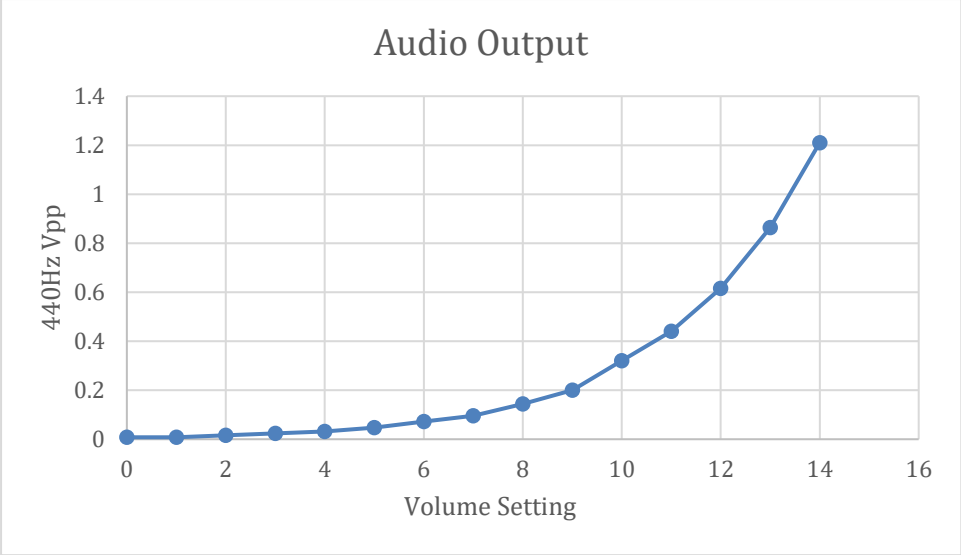


Figure 4 - Mobile Phone characterization – 440Hz mobile phone output voltage vs. volume setting.

Figure 2 above shows that at approximately 70% volume the mobile phone produces an output voltage of approximately **0.32 Vpp**, which was used as the design input signal for the amplifier.

To drive the power amplifier stage effectively, the target output voltage of the first stage was chosen to be approximately **3 Vpp**.

The required voltage gain was therefore

$$A_v = \frac{V_{out}}{V_{in}}$$

Substituting the measured values

$$A_v = \frac{3}{0.32}$$
$$A_v = 9.375$$

The gain expressed in decibels is

$$G = 20\log_{10}(A_v)$$
$$G = 20\log_{10}(9.375)$$
$$G = 19.44\,dB$$

This gain was implemented using the resistor ratio in the operational amplifier feedback network.

The upper cut-off frequency of the filter is given by

$$f_U = \frac{1}{2\pi R_1 C_1}$$

The lower cut-off frequency is given by

$$f_L = \frac{1}{2\pi R_2 C_2}$$

Using the selected resistor and capacitor values allowed the design to achieve a bandwidth suitable for audio signals while maintaining stable amplifier operation.

3.4 Detailed description

The amplifier circuit is divided into several functional sections, including the input stage, bias network, active bandpass filter, power amplifier stage, and output coupling network. Each section performs a specific role in processing the audio signal from the mobile phone input to the loudspeaker output.

3.4.1 Input Stage and Stereo Mixing

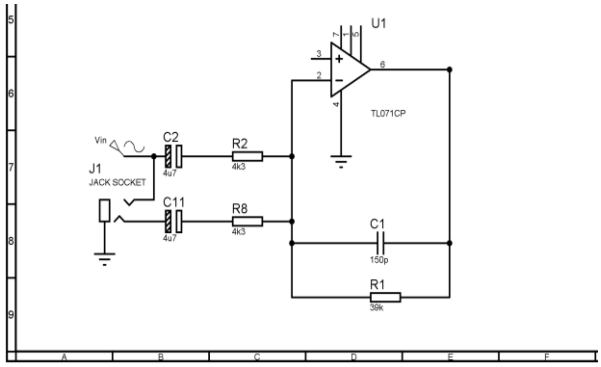


Figure 5 - Circuit sub diagram 1

Figure 3 shows the input stage of the amplifier. The audio signal enters the circuit through a 3.5 mm jack socket connected to a mobile phone audio source. Mobile phone outputs are typically stereo signals consisting of left and right channels.

Resistors R2 and R8 (4.3 kΩ) are used to combine the left and right audio channels into a single mono signal. This resistive mixing network prevents the two channels from directly shorting together while allowing them to be summed safely.

Capacitors C2 and C11 (4.7 µF) are coupling capacitors that block DC components from the input signal while allowing the AC audio signal to pass into the amplifier stage. These capacitors also form part of the high-pass filter that defines the lower cut-off frequency of the amplifier.

Electrolytic capacitors are used in this position because relatively large capacitance values are required to pass low audio frequencies.

3.4.2 Bias Network and Single Supply Operation

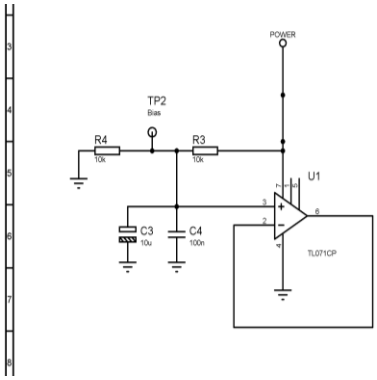


Figure 6 - Circuit sub diagram 2

The circuit operates from a single 12 V power supply, therefore a bias voltage must be created so that the operational amplifier can amplify AC signals without clipping.

Resistors R3 and R4 (10 kΩ) form a voltage divider that generates a midpoint bias voltage of approximately 6 V. This bias voltage is applied to the non-inverting input of the TL071 operational amplifier and shifts the AC signal to the middle of the supply range.

Capacitors C3 (10 µF) and C4 (100 nF) stabilise the bias voltage by filtering noise and ripple from the supply. This ensures the reference voltage remains stable during amplifier operation.

Without this bias network, the amplifier would clip the negative half of the audio signal because the circuit does not have a negative supply rail.

3.4.3 Active Bandpass Filter and Voltage Amplifier

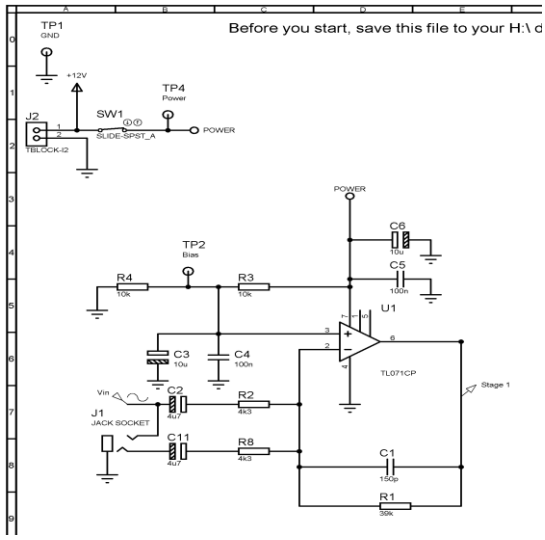


Figure 7 - Circuit sub diagram 3

The first amplification stage is implemented using the TL071 operational amplifier (U1) configured as an active bandpass filter and voltage amplifier.

The voltage gain of this stage is determined by the feedback network consisting of R1 (39 kΩ) and R2 (4.3 kΩ). These resistors set the amplifier gain according to the standard inverting op-amp gain equation.

Capacitor C1 (150 pF) is connected in the feedback path with R1 and forms a low-pass filter that limits the high-frequency response of the amplifier. This defines the upper cut-off frequency of the bandpass filter.

The lower cut-off frequency is determined by the input coupling capacitors C2 and C11 together with the input resistors. These components form a high-pass filter that blocks very low frequencies and DC signals.

Together, these networks produce an active bandpass response, allowing the amplifier to pass audio frequencies while attenuating unwanted low-frequency and high-frequency signals.

3.4.4 Power Supply Decoupling

Capacitors C5 and C6 are placed close to the power pins of the TL071 operational amplifier to provide power supply decoupling.

C5 (100 nF) is a ceramic capacitor that filters high-frequency noise from the power supply, while C6 (10 µF) provides bulk decoupling to stabilise the supply voltage. These capacitors prevent noise and oscillations that may occur due to rapid changes in the amplifier output current.

Decoupling capacitors are commonly placed as close as possible to the integrated circuit supply pins to minimise the impedance of the supply path.

3.4.5 Power Amplifier Stage

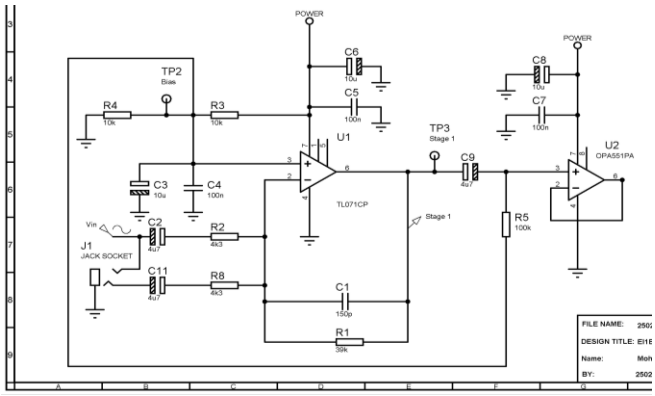


Figure 8 - Circuit sub diagram 4

The second stage of the amplifier uses the OPA551 power operational amplifier (U2). This device provides the higher output current required to drive the loudspeaker.

The OPA551 is configured as a buffer amplifier, meaning it provides current amplification while maintaining approximately unity voltage gain. The voltage gain is therefore primarily determined by the previous TL071 stage.

Resistor R5 (100 kΩ) provides bias stabilisation at the input of the OPA551, ensuring the amplifier operates around the midpoint bias voltage.

Capacitors C7 and C8 provide power supply decoupling for the OPA551 in a similar way to the decoupling network used for the TL071 stage.

3.4.6 Protection Diodes

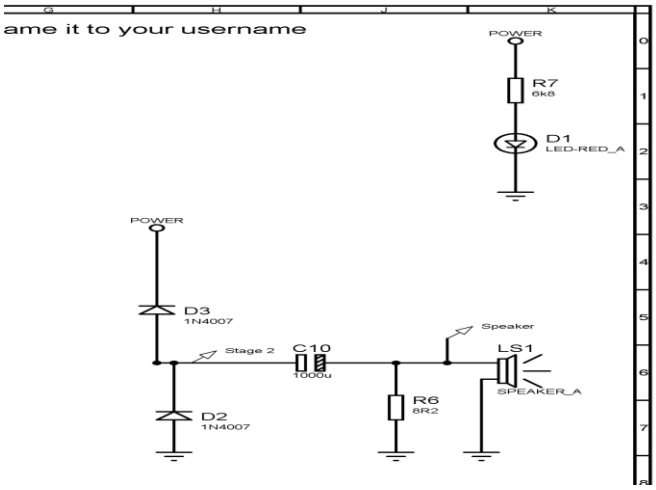


Figure 9 - Circuit sub diagram 5

Diodes D2 and D3 (1N4007) are included in the design to protect the power amplifier from voltage spikes and reverse polarity conditions.

These diodes clamp the output voltage and prevent large voltage excursions that may occur due to inductive loads such as the loudspeaker.

Including protection diodes improves the robustness of the amplifier and reduces the risk of damage to the power amplifier stage.

3.4.7 Output Coupling Network

The amplifier output is connected to the loudspeaker through C10 (1000 µF), which is a large electrolytic capacitor.

This capacitor blocks the DC bias voltage from reaching the speaker while allowing the AC audio signal to pass. Without this capacitor, the DC offset present at the amplifier output could damage the speaker.

The output capacitor, together with the speaker impedance, forms a high-pass filter that determines the low-frequency response of the final output stage.

In simulation, the loudspeaker was represented by an 8.2 Ω resistive load (R6), while the physical design is intended to drive an 8 Ω speaker.

3.4.8 Speaker Load

The output signal drives an 8 Ω loudspeaker (LS1). Loudspeakers typically have low impedance and therefore require significant current to produce audible sound. This is why a power operational amplifier such as the OPA551 is required in the final stage.

During circuit simulation, a resistive load of 8.2 Ω (R6) was used instead of a physical loudspeaker. This value was selected because it is a standard resistor value available in the component library and closely approximates the impedance of an 8 Ω speaker.

Using a resistor instead of a loudspeaker simplifies simulation because a resistor behaves as a purely resistive load, whereas a real loudspeaker is a complex electromechanical device whose impedance varies with frequency due to the inductance of the voice coil and mechanical resonance of the speaker cone.

The small difference between 8 Ω and 8.2 Ω has a negligible effect on the amplifier behaviour. However, using a fixed resistive load allows the amplifier performance, output voltage, and current delivery capability to be analysed more easily during simulation.

3.5 PCB design

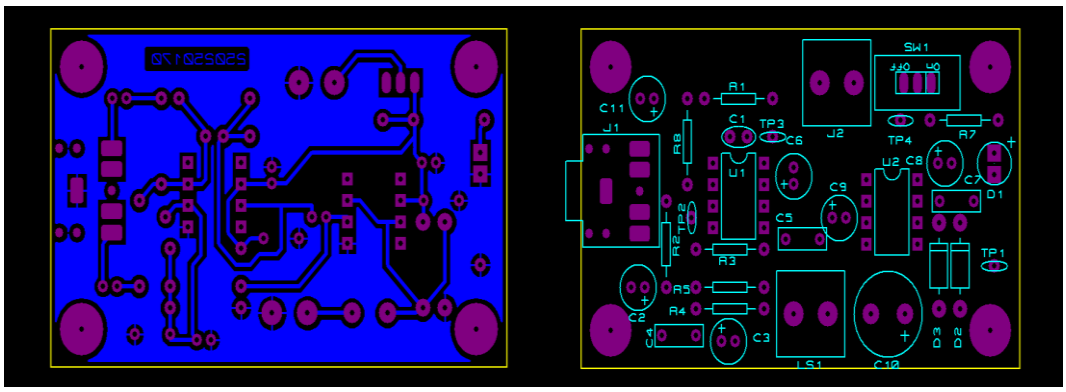


Figure 10 - PCB layout – bottom copper layer of the audio amplifier board (exported from Proteus).

Figure 11 - PCB layout – top copper layer of the audio amplifier board (exported from Proteus).

Figure 8a and Figure 8b show the top and bottom layer layouts of the printed circuit board designed using the Proteus PCB layout tools. The PCB layout implements the schematic design while maintaining clear signal routing and accessible connections for the external components.

The components were arranged to maintain a logical signal flow across the board. The input connectors and signal conditioning components are positioned toward the left side of the PCB, while the power amplification stage and speaker output are positioned toward

the right side. This layout allows the audio signal to move across the board from input to output in a clear and organised manner while reducing the possibility of interference between stages.

External connectors for the power supply, audio input, and speaker output were placed near the edges of the PCB so that wires can be easily connected when the board is mounted on the base plate. This improves accessibility during both assembly and testing.

Power supply decoupling capacitors were placed close to the supply pins of the operational amplifiers to reduce noise on the supply rails and ensure stable amplifier operation. Keeping these capacitors physically close to the integrated circuits minimises the length of the supply path and reduces the likelihood of oscillations.

All copper tracks in the PCB were routed using the T20 track width, which provides sufficient current handling capability for both the signal paths and the speaker output path in this design. Using a consistent track width across the board simplifies routing and maintains uniform electrical characteristics throughout the circuit.

Track routing was designed to keep signal paths short and direct, reducing noise coupling between different sections of the amplifier. Components associated with the same functional stage were placed close together to minimise loop areas and improve circuit stability.

Overall, the PCB layout was designed to follow standard PCB design practices, ensuring reliable operation, good signal integrity, and ease of assembly.

3.6 3D Model from Proteus

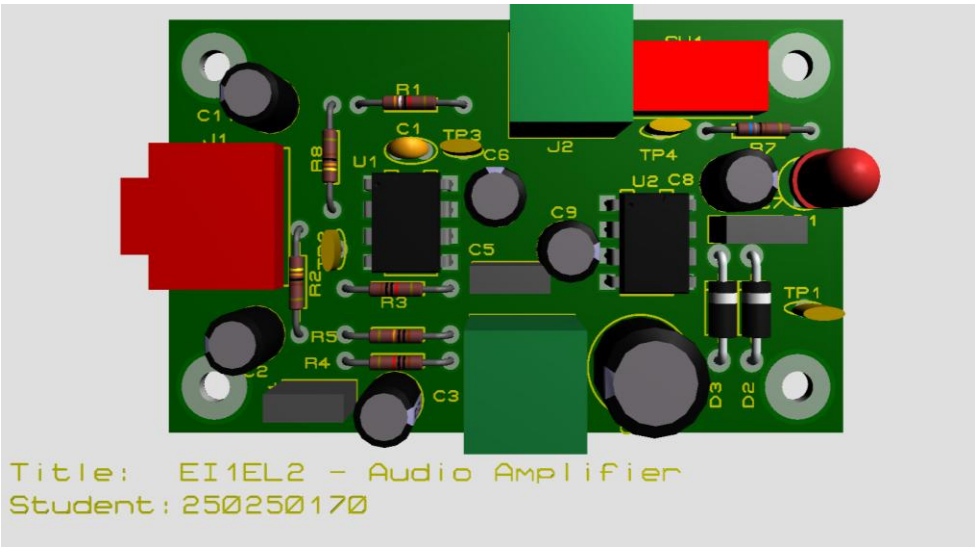


Figure 12 - 3D model of the audio amplifier PCB generated using Proteus.

Figure 9 shows the three-dimensional model of the completed PCB generated using the Proteus 3D visualisation tool. The 3D view provides a representation of how the final assembled board will appear once the components are mounted.

The model illustrates the placement of the operational amplifiers, passive components, connectors, and speaker interface on the board. Component placement was arranged to maintain short signal paths and ensure that connectors remain accessible at the edges of the PCB for external wiring.

The 3D visualisation is useful for verifying component spacing, orientation, and mechanical layout before manufacturing the PCB. This helps identify potential assembly issues and ensures that the final design can be easily mounted onto the base structure of the amplifier.

4. Results

4.1 Prototype testing and results

This section presents the experimental and simulated results obtained during testing of the amplifier. The results are organised according to the two stages of the amplifier: the first-stage TL071 active filter and the complete two-stage amplifier including the OPA551 power stage.

The frequency response of the circuit was measured by applying a sinusoidal input signal and varying the input frequency across the operating range of the amplifier. The output voltage was measured at the relevant test points using an oscilloscope. The gain was then calculated from the ratio of output voltage to input voltage and converted to decibels.

For comparison, simulated results were obtained from Proteus using a frequency response graph under the same conditions as the experimental measurements. The simulated and measured data were exported and plotted in Microsoft Excel so that both responses could be compared on the same graph.

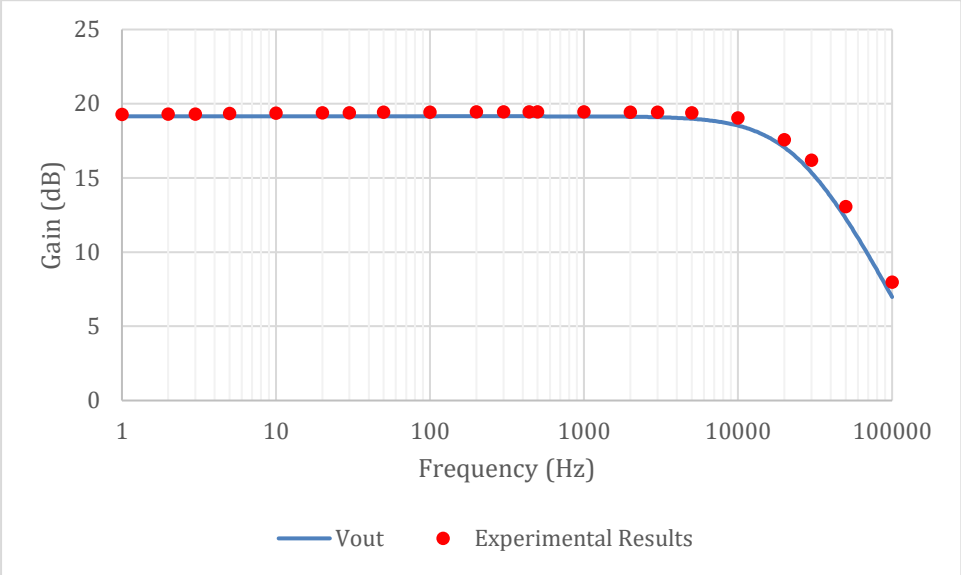


Figure 13 - Simulated and measured frequency response of the amplifier first stage (Breadboard – Dual Power Supply)

Figure 10 shows the simulated and measured frequency response of the first-stage amplifier implemented using the TL071 operational amplifier. The measurements were taken from the output of the first stage while the circuit was powered using a dual power supply during breadboard testing.

The graph shows that the amplifier provides a relatively constant gain of approximately 19 dB across most of the operating bandwidth. At higher frequencies, the gain begins to

decrease due to the low-pass filtering behaviour introduced by the feedback capacitor and the bandwidth limitations of the operational amplifier.

The experimental results closely follow the simulated response across the majority of the frequency range. Small deviations between the simulated and measured results are likely due to component tolerances and measurement uncertainty in the prototype circuit.

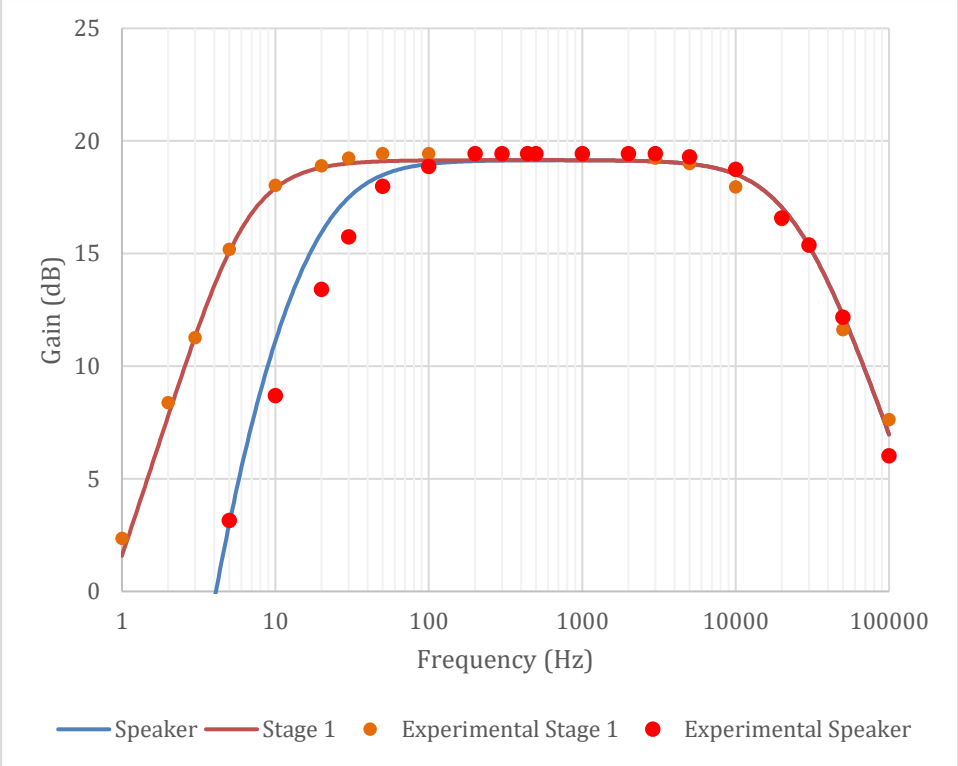


Figure 14 - Simulated and measured frequency response of the two-stage amplifier (Breadboard with Single Power Supply)

Figure 11 shows the frequency response of the complete two-stage amplifier consisting of the TL071 filter stage followed by the OPA551 power amplifier stage operating from a single supply.

The graph compares both simulated and measured responses for the full amplifier. The results demonstrate the expected bandpass response, with attenuation at low frequencies caused by the input coupling capacitors and attenuation at high frequencies caused by the feedback capacitor and amplifier bandwidth limitations.

The measured results show good agreement with the simulated response throughout the majority of the frequency range. Minor differences between the curves may occur due to variations in component values, parasitic effects in the breadboard layout, and differences between the ideal simulation model and the physical components used in the prototype.

Overall, the results confirm that the amplifier achieves the intended frequency response and gain characteristics required for the audio amplification application.

4.2 PCB testing and results

The final amplifier performance was evaluated using the completed PCB assembly. The frequency response of the circuit was measured by applying a sinusoidal input signal from a signal generator through the audio input jack and measuring the output voltage at both the TL071 first stage output and the final speaker output.

During testing, the circuit was powered using a 12 V single supply, as implemented in the PCB design. The output signal was measured using an oscilloscope while the input frequency was varied across the operating range of the amplifier. The gain was calculated from the ratio of the measured output voltage to the input voltage and converted to decibels.

Simulated results were obtained from the Proteus model using a frequency response graph under the same conditions as the physical circuit. The simulated and measured data were exported to Excel and plotted on the same graph for comparison.

Figure 11 shows the simulated and experimental frequency response of the final PCB amplifier. The graph includes the simulated first stage response, measured first stage response, simulated second stage response at the speaker output, and the measured second stage response, including the output coupling capacitor and the 8 Ω load.

The results show that the amplifier achieves an approximately constant mid-band gain close to the designed value of 19 dB. The measured gain closely follows the simulated response within the central portion of the frequency range.

At low frequencies, the gain decreases due to the high-pass filtering effect created by the input coupling capacitors and the output coupling capacitor connected to the speaker load. At high frequencies, the gain decreases due to the low-pass filtering behaviour introduced by the feedback capacitor in the TL071 stage and the bandwidth limitations of the operational amplifiers.

Small differences between the simulated and measured responses are observed, particularly near the lower and upper cut-off frequencies. These differences can be attributed to several factors, including component tolerances, parasitic capacitances present in the PCB layout, and non-ideal behaviour of the operational amplifiers in the physical circuit.

The measured first-stage response is slightly different from the final output response because the second stage introduces additional loading and filtering effects. The output coupling capacitor, together with the 8 Ω speaker load, forms an additional high-pass filter which slightly modifies the low-frequency response of the overall amplifier.

Component tolerances may also contribute to differences between the calculated, simulated, and measured results. Typical resistor tolerances of ±5% and capacitor tolerances of ±10–20% can shift the exact values of the cut-off frequencies. These variations are expected in practical analogue circuits and explain the small deviations observed between theoretical predictions and experimental measurements.

Overall, the measured PCB results show good agreement with the simulation and design calculations, confirming that the amplifier performs as intended within the desired audio bandwidth.

Parameter	Calculation	Simulation	Measured
Gain (dB)	19.44	19.4	19
Lower cut-off frequency (Hz)	8	9	10
Upper cut-off frequency (kHz)	30	28	25
Bandwidth (kHz)	30	27	25

Figure 15 - Calculated, simulated, and experimental frequency response of the final PCB

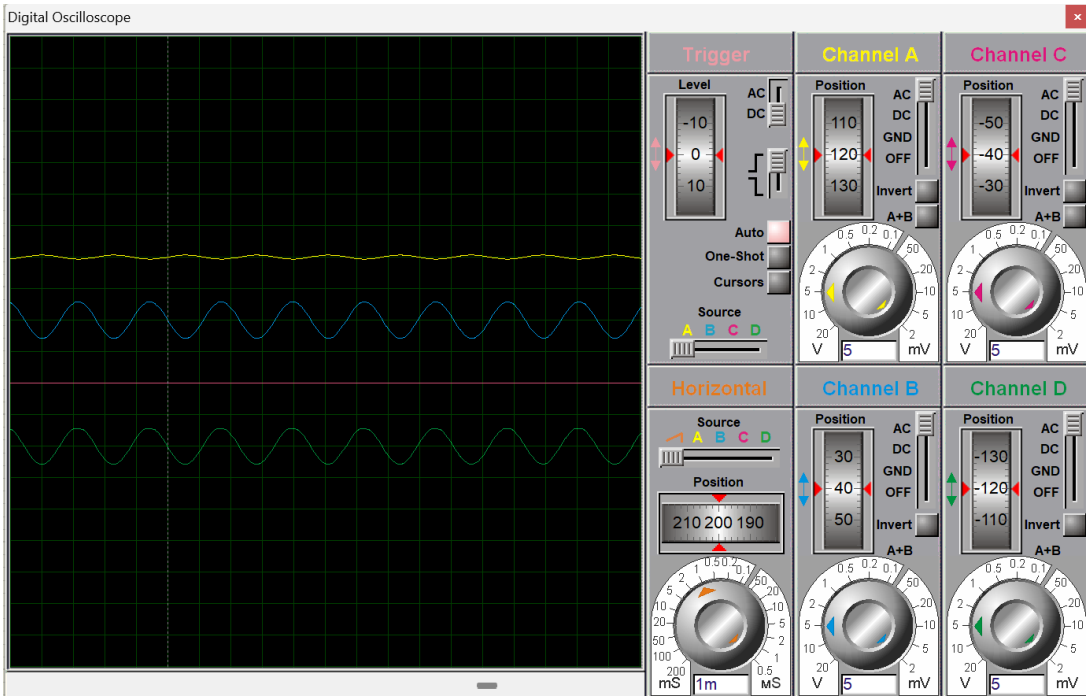


Figure 16 - Input and output 440 Hz sine wave for the two-stage amplifier (final PCB design). Channel A: input signal, Channel B: first stage output (TL071), Channel D: final output at speaker load.

Figure 13 shows the input and output waveforms of the two-stage amplifier when driven by a 440 Hz sinusoidal signal. Channel A shows the input signal, channel B shows the output of the first stage (TL071), and channel D shows the final output at the speaker load.

The input signal has an amplitude of approximately 0.32 Vpp, which corresponds to the design input level derived from the mobile phone output. The signal is a clean sinusoidal waveform at 440 Hz.

The output of the first stage shows an amplified version of the input signal. This waveform is centered around approximately half of the supply voltage, which is expected because the amplifier operates from a single supply and therefore requires a DC bias to allow the AC signal to swing symmetrically.

At the final output, the signal is centered around 0 V because the output coupling capacitor removes the DC bias from the first stage. The measured output amplitude is approximately 3 Vpp, which matches the target output voltage for the amplifier design.

The output waveform remains sinusoidal without visible distortion, indicating that the amplifier is operating within its linear region and that the power amplifier stage is capable of driving the 8 Ω load without clipping.

Overall, the experimental results agree well with the expected behaviour of the circuit and confirm that the amplifier produces the desired gain and output signal level.

4.3 Final assembly

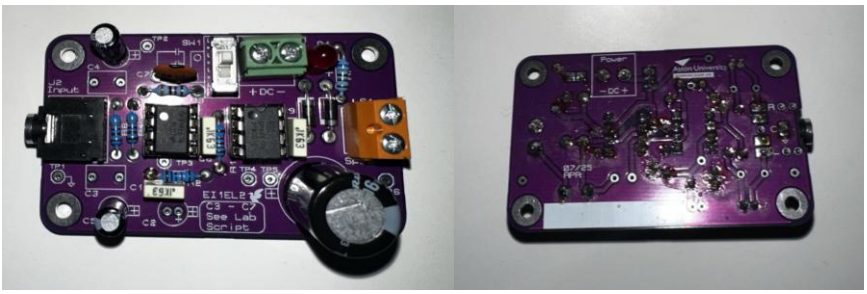


Figure 17 - Photographs of the final assembled PCB amplifier showing the top view with components and connectors, and the bottom view with the copper track routing.

This section presents photographs of the completed PCB amplifier and illustrates the physical layout of the circuit and its external connections. The images show both the top and bottom sides of the assembled board.

Figure 14 shows the final assembled amplifier PCB. The top view illustrates the placement of the main circuit components, including the TL071 operational amplifier used in the active filter stage and the OPA551 power amplifier stage used to drive the speaker load. The audio input is provided through the 3.5 mm audio jack located on the left side of the board, allowing the amplifier to be connected to a mobile phone or other audio source.

A DC power connector is located near the top of the board and provides the single supply voltage required for operation. The power switch and LED indicator allow the amplifier to be easily turned on and provide visual confirmation that the circuit is powered.

The speaker output is connected through the screw terminal on the right side of the board. The large electrolytic capacitor near the output acts as the output coupling capacitor, removing the DC offset from the amplifier output before the signal is delivered to the speaker.

The bottom view of the PCB shows the routed copper tracks that connect the components. These tracks implement the circuit designed in the schematic and allow signals and power to be distributed across the board. The board also includes mounting holes at each corner, allowing the PCB to be securely attached to a base or enclosure.

Overall, the final assembly demonstrates a compact and functional implementation of the two-stage audio amplifier design.

4.4 Evaluation

The aim of this project was to design and implement a two-stage audio amplifier capable of amplifying a mobile phone audio signal and driving an 8 Ω speaker load. Overall, the amplifier operated successfully and achieved the intended design objectives.

The experimental frequency response results shown in Figure 13 and Figure 14 demonstrate that the measured behaviour of the amplifier closely follows the simulated response obtained from the Proteus model. The first stage response shown in Figure 13 maintains an approximately constant mid-band gain of around 19 dB across most of the audio frequency range before decreasing at higher frequencies. The measured data points follow the simulated curve closely, confirming that the TL071 active filter stage performs as expected.

The complete two-stage amplifier response shown in Figure 14 also demonstrates good agreement between simulation and experimental measurements. The graph shows the expected band-pass response of the amplifier, with attenuation at low frequencies due to

the input and output coupling capacitors, and attenuation at higher frequencies due to the feedback network and the bandwidth limitations of the operational amplifiers.

The final PCB performance shown in Figure 15 confirms that the amplifier continues to operate as expected after the circuit was implemented on the printed circuit board. The measured gain remains close to the design value, and the overall frequency response follows the predicted behaviour from both the design calculations and the circuit simulation.

The time-domain behaviour of the amplifier is illustrated in Figure 16, which shows the input and output waveforms of the two-stage amplifier when driven by a 440 Hz sinusoidal signal. The first stage output is biased around approximately half of the supply voltage, which is required for correct operation under single-supply conditions. After the output coupling capacitor, the final output signal is centred around 0 V and produces approximately 3 Vpp across the 8 Ω load. The waveform remains sinusoidal and shows no visible distortion, confirming that the amplifier is operating within its linear region.

Small differences between calculated, simulated, and measured results are expected in practical analogue circuits. These differences are mainly caused by component tolerances, particularly in the resistors and capacitors that determine the cut-off frequencies of the band-pass filter. Parasitic effects from PCB traces, measurement equipment and component non-idealities can also slightly modify the frequency response.

The physical implementation of the circuit is shown in Figure 17, which presents photographs of the final assembled PCB. The board layout successfully organises the amplifier stages and places the input, power and speaker connectors near the edges of the board for easier access. The routing of the tracks on the PCB allows the circuit to operate reliably while maintaining a compact layout.

If the design were to be repeated, several improvements could be made. For example, a larger ground plane could be introduced to further reduce noise and improve signal integrity. The placement of components could also be optimised to shorten critical signal paths, and an adjustable gain or volume control could be added to allow the amplifier output level to be tuned for different audio sources.

Overall, the amplifier design, simulation and experimental testing demonstrate that the circuit performs as intended and successfully meets the requirements of the project.

5. References

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6. Appendices

6.1 Bill of Materials (BoM)

Bill Of Materials for EI1EL2 - Audio Amplifier

Design Title	EI1EL2 - Audio Amplifier
Author	250250170
Document Number	Mohammed, Sabir
Revision	1.0
Design Created	11 March 2026
Design Last Modified	11 March 2026
Total Parts In Design	32

Category	Quantity	References	Value	Stock Code	Unit Cost
Capacitors	1	C1	150p		
Capacitors	3	C2,C9,C11	4u7		
Capacitors	3	C3,C6,C8	10u		
Capacitors	3	C4-C5,C7	100n		
Capacitors	1	C10	1000u		
Resistors	1	R1	39k		
Resistors	2	R2,R8	4k3		
Resistors	2	R3-R4	10k		
Resistors	1	R5	100k		
Resistors	1	R6	8R2		
Resistors	1	R7	6k8		
Integrated Circuits	1	U1	TL071CP		
Integrated Circuits	1	U2	OPA551PA		
Diodes	1	D1	LED-RED_A		
Diodes	2	D2-D3	1N4007		
Miscellaneous	1	J1	JACK SOCKET		
Miscellaneous	1	J2	TBLOCK-I2		

Miscellaneous	1	LS1	SPEAKER_A
Miscellaneous	1	SW1	SLIDE-SPST_A
Miscellaneous	1	TP1	GND
Miscellaneous	1	TP2	Bias
Miscellaneous	1	TP3	Stage 1
Miscellaneous	1	TP4	Power

Table 1 - Bill of Materials listing all components used in the final PCB amplifier design

6.2 Full circuit design

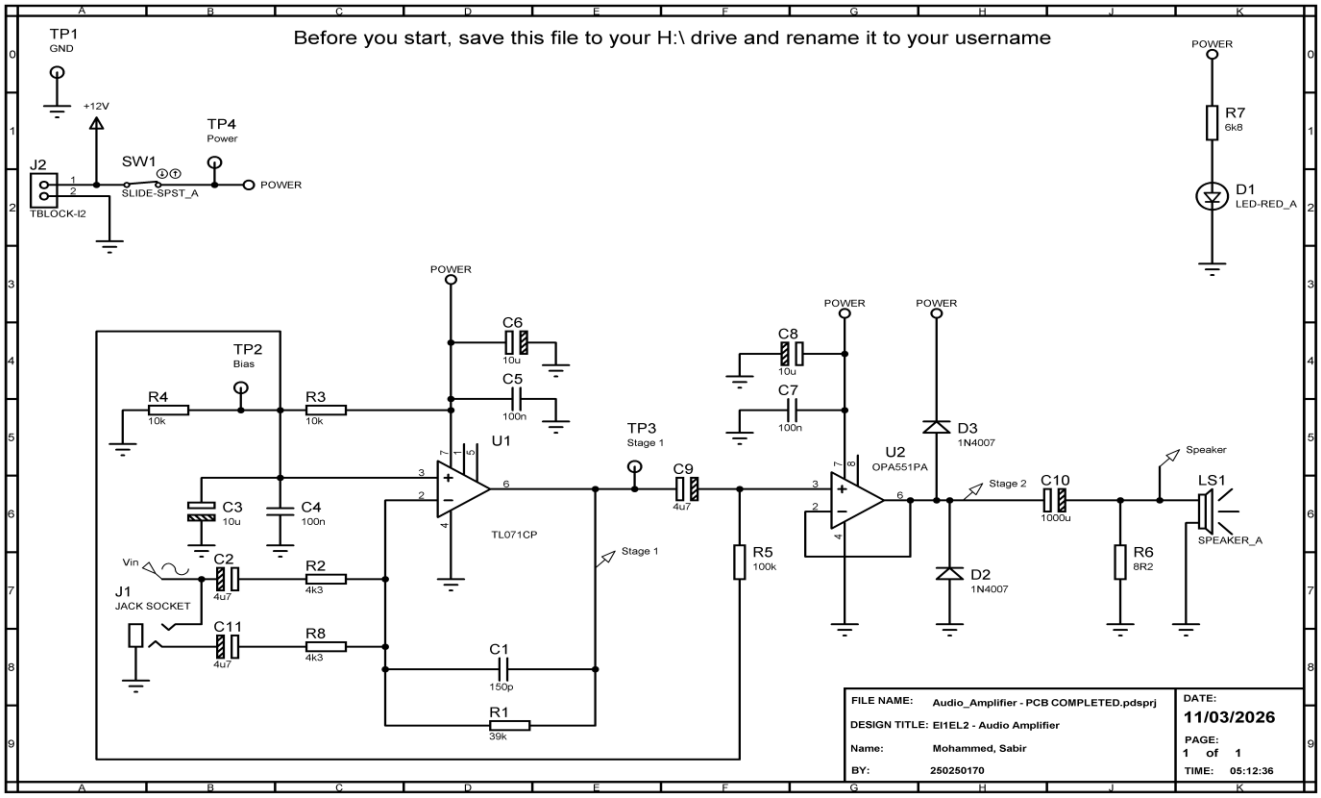


Figure 18 - Full Proteus circuit diagram